

Ameya Deshmukh

Boston, MA · +1 857-396-7408 · deshmukh.amey@northeastern.edu · [LinkedIn](#) · [GitHub](#) · [Portfolio](#)

TECHNICAL SKILLS

Programming: Python (Pandas, NumPy, Scikit-Learn, PyTorch) | SQL | R
GenAI & Agentic: RAG | Agentic Architectures | Multi-Agent Systems | LLMs | LangChain | Prompt Engineering
ML & Statistics: Classification and Regression Pipelines | k-means Clustering | Feature Engineering | Statistical Analysis | NLP
Evaluation: MRR and Hit Rate | Evaluation Harnesses | Adversarial Test Cases | Embedding Models
Production Engineering: MLflow | Docker | CI/CD | REST APIs | Git | pytest
Data & Cloud: ETL Pipelines | Databricks | Redshift | Tableau | AWS | Azure | GCP

EDUCATION

Master of Science in Data Analytics Engineering Dec 2025
Northeastern University, Boston, MA | GPA: 3.75/4.0

Bachelor of Engineering in Electronics & Telecommunication May 2022
Vishwakarma Institute of Information Technology, Pune, India | GPA: 3.76/4.0

WORK EXPERIENCE

AI Engineer Jan 2026 – Present
Humanitarians AI *Boston, MA*

- Run granular segment-level analyses and outlier detection to isolate key performance drivers, writing step-by-step methodology documentation reviewers can independently verify.
- Design adversarial test cases and evaluation harnesses that surface where models produce confident but unsupported answers, documenting each failure alongside a verified correct result.
- Prioritize across parallel technical and strategic projects with limited supervision, taking work from ambiguous business requirement to running solution.

AI Engineer Jan 2025 – July 2025
Boehringer Ingelheim *Ridgefield, CT*

- Engineered **PySpark** and **Pandas** feature pipelines in **Databricks** processing **2M+** records from **Amazon Redshift**, cleaning and standardizing source data to improve upstream data quality **40%**.
- Deployed **MLflow** and **Docker MLOps** workflows with **CI/CD** integration, monitoring production model performance against **SME**-validated accuracy benchmarks and presenting insights via **Dash** dashboards.
- Architected **LangChain** multi-agent system with Router, Supervisor, and Collaborative patterns, implementing **ChromaDB** vector search with k-means centroid routing.
- Built **PyMuPDF** OCR document intelligence extracting unstructured records into **HIPAA**-compliant retrieval infrastructure hosted on **Azure** cloud.
- Facilitated requirement intake sessions with business stakeholders to scope **AI** use cases, defining success metrics and translating results for non-technical audiences.

Data Scientist May 2024 – Nov 2024
Northeastern University *Boston, MA*

- Engineered **Scikit-Learn** classification and regression pipelines across **50K+** records on **GCP**, delivering **SQL**-driven feature stores that cut training time **35%**.
- Delivered **Flask REST APIs** and a **React** dashboard enabling real-time access for **3 faculty teams**, reducing ad hoc reporting requests **40%**.
- Mentored student engineers through code review against documented standards, raising the quality bar across all of the team's delivered project output.

Data Analyst Jan 2023 – May 2023
Newtuple Technologies *Remote*

- Built an **NL-to-SQL** agent using **LangChain** and prompt engineering converting natural language into queries, reducing analyst data access time **30%**.
- Designed **dbt** data models in **PostgreSQL** and automated **ETL** pipelines on **AWS**, improving business **KPIs 12%** through targeted feature engineering.
- Profiled and cleaned **500K+** records using **Python** and **SQL**, building **Tableau** dashboards that informed product decisions.

PROJECTS & APPLIED RESEARCH

Production RAG on Financial Disclosures | Python, LangChain, Sentence-Transformers | [Link](#) 2026

- Built a **600**-document retrieval system with an **MRR** and hit-rate eval harness, benchmarking **3** embedding models across quality, index build time, latency, and storage.

Observable Agent System | Python, LangGraph, Opik, Pytest | [Link](#) 2025

- Built a **LangGraph** agent with a retry cycle gated by a zero-**LLM** fabrication validator enforcing provenance and metric thresholds, traced in **Opik** with **32** tests.

Multi-Agent RAG Research Assistant | Python, LangChain, ChromaDB, Streamlit | [Link](#) 2024

- Architected multi-domain agentic research system using **LangChain** with centroid-routed **RAG** pipeline, **SQLite** memory, and **LLM**-as-judge evaluation framework deployed on **HuggingFace**.